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Research on Personalized Search Based on ElasticSearch

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***Abstract*—In the context of today's big data era, with the rapid development of information technology in all walks of life, information data has exploded. People have been used to relying on the Internet to obtain effective information, but massive data is easy to cause information overload, how to quickly and efficiently obtain useful information for users has become the main problem facing search engines. In order to better study the ElasticSearch application in personalized search, this paper proposes a personalized search engine based on ElasticSearch, on the basis of improve the retrieval efficiency and retrieval accuracy, analysis of user search history records and utilization, let the search engine can be more intelligent, friendlier interact with the user, and makes search results more accord with the actual needs of users. Then, this paper tests and analyzes the search engine in many aspects. The test results show that the search rate has increased by 32.26%, which verifies the practicability, effectiveness and real-time performance of the search engine.**

Keywords—Grey Relational Clustering, User Preference, Vertical Search, Personalized Sorting.

I. Introduction

The scope of information retrieval changed from the traditional file system to Internet resources. With the research and development of Internet information resource retrieval technology, search engine technology emerged [1-2]. At present, search engines can collect information through the virtual world of the Internet, but also can actively build a platform to digitize the real world information. Search engine is becoming the largest information gathering and distributing platform in the history of mankind. Since there is a search engine, the way people get information and knowledge is much simpler than before [3-4].

Conditional preference network (CP-NET) is widely used to express qualitative preference. Because users are sometimes unwilling or unable to specify a full CP-NET, this makes personalized search ineffective. Hongbing Wang proposes a method to perform personalized search using an incomplete CP-NET. In order to improve the accuracy of personalized search, a preference recommendation scheme was studied to supplement users' CP-NET. Hongbing Wang conducted extensive simulation and user research to prove the effectiveness of the method [5]. Interactive evolutionary algorithm is applied to personalized search to reduce user fatigue and improve search efficiency. On this basis, Yang Chen proposed a fast interactive estimation distribution algorithm based on personalized search domain knowledge. First, from the social knowledge of personalized search, a Bayesian model describing the variable preference distribution of new users is deduced. Then, the model is used to improve the performance of the algorithm in two aspects [6]. Dong Zhou proposed a new model to construct rich user profiles using external corpus for personalized search extension. Dong Zhou's model integrates the current state of the art text representation learning framework (called word embedding) with the topic models in two sets of pseudo-aligned documents. Based on user profile, two new query extension technologies are established [7].

This article designs and implements a distributed search engine based on ElasticSearch in the video domain. Firstly, this paper expounds the research background and significance of the project [8]. Then, this paper introduces the technology involved in the implementation of the system. Secondly, this paper expounds the functional and non-functional requirements, outline design, detailed design and implementation of the system. Finally, the system is tested and its performance is analyzed [9-10].

II. Personalized Search Based on ElasticSearch

A. ElasticSearch

ElasticSearch provides an easy-to-use and powerful API to interact with the outside world. ElasticSearch supports Java, Python, C#, PHP, and other mainstream languages. The nodes within the cluster communicate through apis. ElasticSearch'S RESTFul. API supports indexing, searching, deleting and other operations directly as HTTP requests, with documents represented as JSON during indexing and searching.

ElasticSearch default document sorting USES the TF-IDF model to calculate ElasticSearch. In the TF-IDF model, there are two factors that affect the ordering of correlation: Term Frequency (TF) and Inverse Document Frequency (IDF), which respectively represent the occurrence Frequency of a certain word in the Document and the number of documents in the Document set. The higher the word frequency, the higher the relevance. The more documents, the lower the relevance:

$$score(q,d)=\sum_{t}^{q}\sqrt{TF_{t,d}}*IDF_{t,d}^{2}*norm(t,d)*boost(t) \quad (1)$$

Score is a relevancy score. The Norm is a regular value for the length of a field; the shorter the field, the higher the weight. Boost silent is 1, so the user can assign weights.

B. Personalized Search

As a search system, through the establishment of a user's click behavior such as data analysis to determine the user's interest preference, this method is based on the user's own historical data in the user recommendation, but the user own historical data sparse cannot solve the user's intent related

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queries, so relevant to mining, according to the user to solve the problem of sparse similarity degree.

The method of user similarity calculation is introduced before the classification of recommendation algorithm, because the calculation of similarity is the key step of collaborative filtering algorithm. The selection of similarity method determines the accuracy of final recommendation results, calculates the degree of similarity between users and products, and recommends the most similar products to users. Where, the utility function U (c,s) is defined as follows:

$$u(c,s) = score(userprofile(c), content(s)) \quad (2)$$

III. Design of Simulation Experiment

A. Experimental Background

In recent years, with the rapid growth of the number of search engine users and the utilization rate of Internet users, the general search engine ushered in a new stage of rapid development, in this stage must also produce some corresponding problems, for the return of comprehensive search results of the general search engine, will lead to the decline in the accuracy of search results. Such problems promote the generation and development of personalized search engines. Personalized search is relative to general search, it has its own unique advantages, such as: high accuracy, in-depth search results, etc., it only applies to the user based on the personalized perspective of search, limited the search range, improve the search depth.

B. Experimental Design

This paper is based on the personalized search system of online shopping platform. Therefore, the data set adopted in this chapter is trained by a shopping website which provides all transaction and user record data of 2158 users and the quantity of daily commodities of 5167.

When Nutch collects large data sources, it needs to be integrated into the Hadoop environment to complete the distributed collection task. The hardware layout requires two servers with Linux installation, whose names are set as Master and Slaver respectively. The login username is set as Nutcher, and the same information is configured in the etc/hosts file, thus completing the configuration of the hardware. Kibana is an open source browser based on analysis and search ElasticSearch dashboards. The version used this time is Kibana-4.0.1. Download the pre-compiled Kibana-4.0.1 from the ElasticSearch official website via WGET and unzip it to the directory to run directly. Some experimental results are shown in Table I.

TABLE I. Experimental results

Search keywords	Quantity of results (per bar)	Time consuming (per second)
Shanghai	13919	0.03
Children's clothes	2745	0.1
Fast food	830	0.04
Frozen food	731	0.08
Cold chain transportation	1072	0.12
The average	3859.4	0.074

IV. Performance Analysis of Personalized Search Based on ElasticSearch

A. Research and Analysis Based on ElasticSearch in Personalized Search

As shown in Figure 1, the tests were performed in the following aspects: index segmentation, query segmentation, professional term segmentation, search result comparison, and query speed testing. In terms of index segmentation, the search engine can perform fine particle segmentation to ensure the recall rate. In terms of query segmentation, it can perform maximum word length segmentation to ensure the accuracy. In terms of professional vocabulary segmentation, can make accurate recognition and segmentation of professional vocabulary; In the search results, shopping information search more professional; In the query speed, has the good search speed. The ApacheBench performance test for the project was used to measure the maximum performance that the back-end server could provide, in particular the amount of traffic that the back-end server could carry per second. The total number of test requests is 10000, the single concurrent number is 1000, the server can carry 1100 requests per second, each request processing time is 908ms. Take the key word "children's wear" for example, the total number of tests is 1000 times, and the number of concurrent tests is 100 times. The server can carry 74 requests per second, and the processing time of each request is 1349ms.

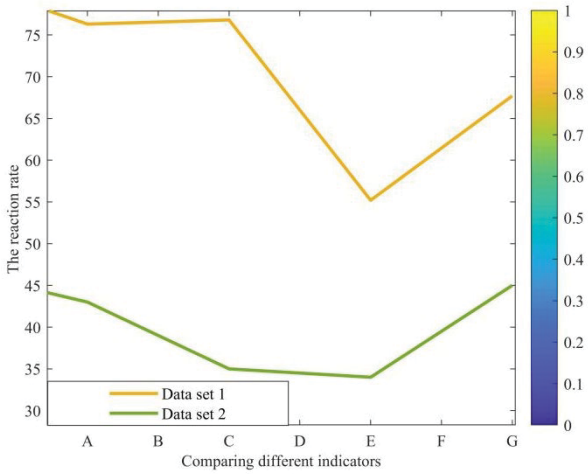

Fig. 1. Comparison of test rates from different angles

In the data, there are 4 18-dimension user features and 5 23-dimension commodity features. After dimension reduction, there are 46 user behavior features, namely 46 dimensions. The network input layer of each commodity is set as 87 and the output layer as 1 output. During the experiment, the first 10 records (de-weighting) of the search results were selected for each query as experimental data for analysis. This experiment mainly observed the change of the ranking of search results before and after adding the algorithm based on user interest. The reordering algorithm based on user's interest proposed in this paper requires the search results to be sorted twice, the first time according to the relevancy of the search results, and the second time according to the user's interest based on the first sort. As the number of results returned by search engines increases, the query accuracy of search engines gradually decreases, and the reordering accuracy based on users' interest is higher than that of traditional relevancy. Therefore, the reordering based on users' interest has a higher accuracy, and the query results are more in line with the real

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query needs of users.

As shown in Figure 2, the average absolute error and the mean square error than other algorithms, the traditional search engine algorithm is lower, but the error curve is relatively large fluctuations in performance is not stable, and the use of ElasticSearch after optimization, besides the low error than traditional algorithm and error curve in comparison is more stable, shows that ElasticSearch effect is more significant in terms of simulated user preference model. At present, the similarity between users computing method is more, but the traditional calculation, cosine similarity and Pearson correlation coefficient, the common characteristic is based on the vector similarity calculation, belongs to the strict matching between objects, there must be a common commodity score for similarity is more accurate, so the data cannot provide credible data sparseness similarity. Since this chapter is aimed at the problem of sparse user data, the grey correlation clustering algorithm is adopted to analyze the correlation between various factors of the system to calculate the similarity between users.

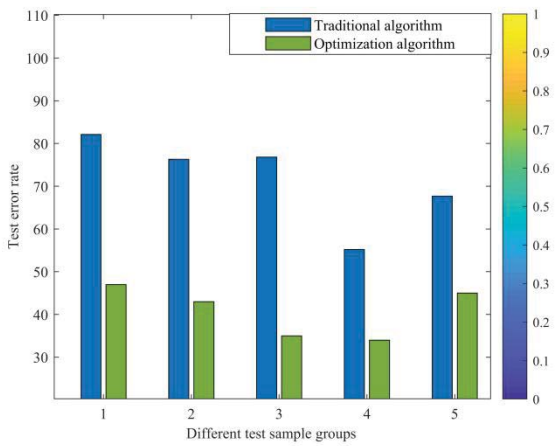

Fig. 2. Comparison of error rates between traditional and optimized algorithms

As the number of samples increases, the average absolute error decreases gradually, and the rate of error decline obtained from the slope of tangent line also decreases. Since the number of recent neighbors indicates an increase in the number of similar users, the weighted results will better fit the real interests of the users, so the recommendation results will be more accurate and the MAE curve will become flatter. The COLLABORATIVE filtering MAE based on grey correlation aggregation is always lower than other similarity analysis methods, so this method improves the recommendation effect for users. In ROC comparison, both curves show a rising trend with the increase of the number of the nearest neighbor, but the increasing speed also gradually slows down. The accuracy of each search engine can be calculated based on the correct total number of search results related to the course. A total of 50 search results were selected in the whole test. After calculation, the accuracy of Google search was 30%, that of sogou search was 20%, that of Bing search was 28%, and that of this system was 74%. Through comparison test, the search accuracy of personalized search engine system in online shopping category is higher than that of other general search engines. The search engine in the field of professional information search has more advantages, search accuracy is higher, the search results of the ranking more user satisfaction.

B. Research Recommendations Based on ElasticSearch in Personalized Search

The user privacy protection model based on location services consists of three types of entities, namely data owner, user and cloud server. The data owner is responsible for generating the Key SK and sending it and the file ciphertext decryption Key to the user through a secure channel. He is also responsible for building the searchable index of the file and uploading the searchable index I and the encrypted file C to the cloud server. On the client side, the user constructs the user model by means of location transformation, then generates the user query, and sends the encrypted query and parameter K to the cloud server. After the cloud server gets the user's search request, it calculates the score of the user query and each file index according to the security inner product, and returns the top K files with the highest score to the user. This method not only protects the user's privacy but also ensures the availability and accuracy of the user's location information by means of location conversion and encryption, avoids the disadvantages of the need for trusted anonymity in location-based services, and at the same time realizes the user's requirements for personalized search and privacy protection. At the same time, this method can greatly save the communication cost and improve the user experience.

Through the training of the proposed algorithm will users - commodity grade matrix, and learn to not buy goods training predict the user's interest degree, fill in the missing user - goods score matrix, form a complete user - goods rating data, so that it can be for users to search out all the goods ordered by according to the score. At the same time, filling the predicted score also increases the data density, so that for new users, there is no missing item in the score matrix of the nearest neighbor set when calculating the similarity of users, which makes the calculated user score matrix more accurate. Compared to traditional training methods, methods such as average value or use the default for missing items, although relatively simplified calculation and methods, but after all the user's score is not the same to the default goods, such as no contact for two pieces of goods, our evaluation is bound to be somewhat deviation, so as the value of the training process will fill their scores is unable to truly reflect the user preferences.

V. Conclusions

Based on ElasticSearch studied the personalized search, according to the features of users and commodities established search engines, the procurement of users based on the online platform to provide historical data, given the large number of characteristic collection first dimension reduction processing, re-use ElasticSearch related algorithm for training, build a search engine users - goods, and the forecast of goods not to score project, through experiment, can accurately locate the user interest preferences.

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